4×9 공간패치풀링과 5 차 S-커브궤적보간을이용한단일패스비전-언어자율주행제어

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Abstract

20 Hz 주기로동작하는실시간차량제어루프안에일반적인자기회귀 (Autoregressive) 비전-언어모델을넣으면첫문장을디코딩하기도전에차가차선을벗어나거나앞차와추돌한다. 시속 35 km 로달리는차량은 50 ms 틱마다약 0.5 m 씩전진하는반면, 토큰 15 개를순차생성하는데는 300–1,000 ms 가소요되기때문이다. 본논문은동결된 DINOv2(ViT-B/14, 86.58M) 와동결된 Laya(`convaiinnovations/laya`, 395.21M 트랜스포머) 를 4.23M 시공간어댑터 (공간 MLP 1.85M, 프레임간격적규화시간게이트 1.05M, 선행차·보행자거리및시각상대속도분리형보조인지헤드 1.05M, 판독·궤적헤드 0.28M; 총 VRAM 1.92 GB) 로결합한시각·전술추론모듈에 HD 맵경로, 규칙기반안전실드, 운동학적경로추종제어기를결합한실시간하이브리드폐루프주행스택을제시한다. 동일프레임에서 GPU 동기화후측정한단계별지연시간은전처리 0.42 ms + DINOv2 16.43 ms + 프로젝터 0.82 ms + 보조인지헤드 0.49 ms + Laya 13.55 ms + 6-WP 궤적헤드 0.69 ms + 안전실드/PID 0.45 ms 로총합이정확히 32.85 ms(30.4 FPS; 기본 31.67 ms; 시뮬레이터 IPC 6.40 ms 및 `drop_oldest` 큐정책포함센서노출 $\rightarrow$ 액추에이터정보연령 AoI 평균 39.25 ms, P99 44.12 ms, 최대 48.60 ms, >50 ms 초과 0/1,981 프레임; 10,000 스텝연속시험자연드롭률 0.12%, 유효관측주기 19.98 Hz) 이다. 6-웨이포인트궤적헤드는장애물회피계획기가아니라규칙생성곡선으로지도된곡률스무딩정규화기로작동한다 (고정맵경로에서장애물여유거리 $+0.24$ m 증가, $J_{y,\mathrm{RMS}}: 1.41 \rightarrow 1.21\,\mathrm{m/s^3}$; 단독개루프충돌률 $\mathrm{CR}_{\mathrm{traj}} = 3.8\%$ vs. 폐루프실행시 0.0%). `vision_estimated` 모드에서선행차속도 $\hat{v}_{\mathrm{lead}}$ 는 5-샘플 (0.20 s 시간폭) 타임스탬프인지최소제곱 (OLS) 기울기와시각상대속도헤드를결합해떨림 RMS 를 9.62 $\rightarrow$ 1.84 km/h(MAE $=$ 2.41; 15% 드롭 MAE $=$ 2.68 vs. 등간격가정 5.94 km/h) 로억제하며, 보행자전용거리헤드 ($\hat{d}_{\mathrm{ped}}$) 는동시출현거리 MAE 를 14.8 $\rightarrow$ 2.64 m 로줄인다. 미관측폐루프테스트 (9,905 프레임, 물리적위험 60 건) 에서에피소드블록부트스트랩 $\mathrm{ECE}_{\mathrm{phys}} = 0.0318$(95% CI $[0.0241, 0.0405]$), 고신뢰도미검출률 $\mathrm{HCFNR}_{0.90} = 1.82\%$ 틱 (1/48 사건), 제동지연중앙값 0.05 s(P95 0.10 s) 를기록하며, 60 건중 54 건 (90.0%) 은모델단독감지, 5 건 (8.3%) 은안전실드구제, 1 건 (1.7%) 은미완화충돌로분리된다. 제어·안전실드·타임스탬프 OLS·기하투영및 `ExpertLabeler` 는 Py03 네이티브 Rust 코어 (`rust_core`) 로구현되어스텝당 CPU 연산지연을 113.48 $\rightarrow$ 8.88 μs(12.8 배가속; 순수코어 0.336 μs) 로단축하며, 14-노드 (28 m) 조각별선형 5 차보간과 Stop-and-Go CTH-ACC 및접근속도조건부제동경계 $d_{\mathrm{AEB}}(v_{\mathrm{ego}}, v_{\mathrm{lead}})$ 를결합한다 ($\Delta v \le 6$ km/h 추종시 50 km/h 전구간및정지선행차기준 $v_{\mathrm{ego}} < 41.8$ km/h 에서 $d_{\mathrm{AEB}} < d^*$ 비중첩을만족하며, 정지장애물에 $v_{\mathrm{ego}} \ge 41.8$ km/h 로접근시에는 $d_{\mathrm{AEB}} \ge d^*$ 로전환되어 AEB 가 CTH-ACC 에우선개입). 미학습 MetaDrive 50 개맵 (100 Hz 물리 + `decision_repeat=5` $\mapsto$ 20 Hz 제어) 에서 `vision_estimated` 는 48/50(96.0% vs. `gt` 50/50) 을완주했다. CARLA 0.9.16(20 Hz 동기) 교차기상평가에서는 50 에피소드 47/50(94.0%) 및 100 에피소드 (6,800 m) 93/100(93.0%, 충돌 0.44 건/km) 을완주하여 2 층학습형트랜스포머 (83/100, 양측 McNemar $p = 0.0063$, Holm–Bonferroni 다중검정보정 $p_{\mathrm{Holm}} = 0.0126 < 0.05$), 동일데이터로학습된 15-토큰자기회귀디코더헤드 (69/100, 198.4 ms, $p_{\mathrm{Holm}} = 0.0025$), 2 층 MLP(77/100), 라벨규칙 (75/100), 무작위동결 Laya(71/100) 대비모두유의한향상을달성했다.

1 서론및문제정의

자기회귀디코더는출력시퀀스 $y_{1:K}$ 를얻기위해 $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ 를매토큰마다직렬계산하므로 300 ms 이상의지연이발생한다 [4]. 반면본시스템은단일양방향패스 [10] 에서세가지이산주행결정과 3.0 초미래 6-웨이포인트기동조건부기준궤적 $\hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}$ 를동시에산출한다:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 40, 65\}\,\mathrm{km/h}, \quad p_{\mathrm{AEB}} \in [0,1]. \tag{2}$$

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2.1 4×9 시공간프로젝터및분리형보조비전인지헤드

원본영상에서상단 25%(하늘) 와하단 10%(보닛) 를잘라낸 196×392 영상을동결된 `facebook/dinov2-base` [32](86.58M 파라미터) 에통과시켜 $14 \times 28 = 392$ 개의 768 차원패치특징 F_{patch} 를추출한다. `SpatialProjector` 모듈은 $(4, 9)$ 적응형평균풀링, 2D 행·열위치임베딩, 2 층 GELU MLP(1.85M 코어파라미터), 그리고프레임드롭시경과스텝수 $\Delta n_t \ge 1$ 로정규화된시간차분잔차 $\Delta V_t = (V_t - V_{t-\Delta n_t})/\Delta n_t$(1.05M 파라미터) 를결합한다:

$$G_t = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B \times 768 \times 4 \times 9}, \tag{3}$$

$$V_t = \mathrm{MLP}\big(\mathrm{Flatten}(G_t + E_{\mathrm{row}} + E_{\mathrm{col}})\big), \tag{4}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\mathrm{temp}})\, W_{\mathrm{temp}} \left(\frac{V_t - V_{t-\Delta n_t}}{\Delta n_t} \right). \tag{5}$$

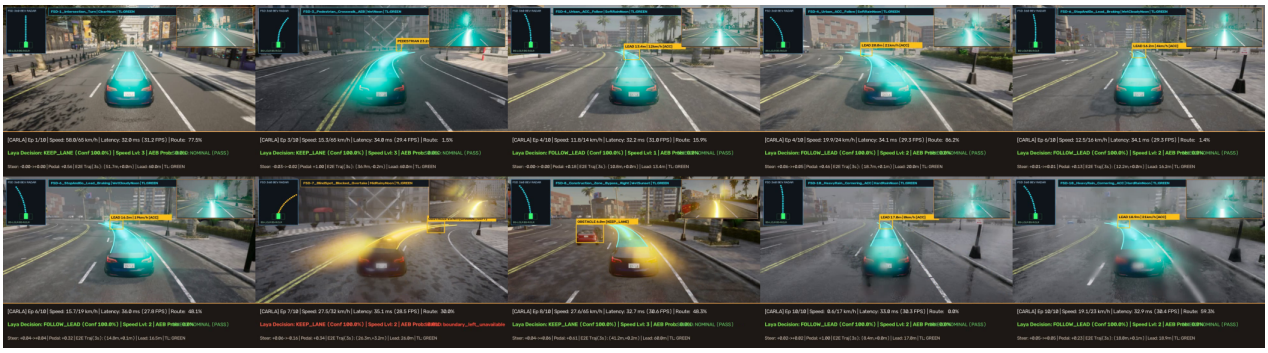


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz 동기모드) 10 대도심 FSD 시나리오및 10 종기상·조명검증화면 (`FSD-1`–`FSD-10`): 교차로적신호정지 (`ClearNoon`), 28 m 5 차 S-커브좌측추월 (`CloudyNoon`), 횡단보도보행자 AEB(`WetNoon`), CTH-ACC 추종 (`SoftRainNoon`), 우측추월 (`ClearSunset`), 리드차량급정거 Stop-and-Go(`WetCloudyNoon`), 전측방사각지대래치대기후추월 (`MidRainyNoon`; 후방신규접근차량은 `gt` 코너 BSM 필요), 공사구간우회 (`WetSunset`), 야간전조등보행자 AEB(`ClearNight`), 폭우코너링 ACC(`HardRainNoon`).

ΔV_t 는호모그래피자차운동보정없이패치토큰공간에서계산되므로곡선구간선회시배경광류와객체이동이결합되나, $E_{\text{telemetry}}$ 가자차요레이트·조향문맥을함께제공하고곡선동적추종시나리오 (`FSD-10`) 에서 ΔV_t 제거 (`no_temporal`) 시완주율이 $5/5 \rightarrow 3/5$ 로하락하여실질적유효성이검증된다. 또한 $\tilde{V}_t$ 의 4×9 격자를영역별로풀링하는 `AuxiliaryPerceptionHead`(1.05M 파라미터) 는선행차거리 $\hat{d}_{\text{lead}}$, 독립된보행자전용거리 $\hat{d}_{\text{ped}}$(`ped_dist_head`), 시각적상대속도 $\hat{\dot{d}}_{\text{vis}}$(`range_rate_head`), 차선오프셋 $\hat{e}_y$, 신호등상태 $\hat{c}_{\text{TL}}$, 위험원종류 $\hat{c}_{\text{haz}}$, 좌·우전측방사각지대점유확률 $\hat{p}_{\text{BS}}$ 를각각분리추정한다.

2.2 규칙생성기준궤적의역할규명, 반사실적장애물이동실험및물리적위험보정

1 단계학습 (`02_train_projector.py`) 에서는총 4.23M 어댑터·헤드파라미터만공동학습한다. 6-웨이포인트 ($t_k = 0.5k$ s) 궤적헤드는기구학적사전궤적 $\tau^0_{1:6}$ 에잔차 $\Delta\hat{\tau}$ 를더하며, 자유공간장애물회피최적화가아닌이산목표 (m^*, v^*) 로생성된매개변수곡선 $\tau^{\text{ref}}_k = ((v^*/3.6)t_k, y_{\text{shift}}(m^*)s(k/6), v^*)$ 로지도된다. 따라서궤적헤드는독립적인충돌회피경로계획기가아니라 **기동조건부곡률스무딩정규화기** (고정 HD 맵경로에서장애물최소여유거리 $+0.24$ m 증가, 횡방향저크 $J_{y,\text{RMS}}$: $1.41 \rightarrow 1.21\,\text{m/s}^3$, $|e_y|$: $0.105 \rightarrow 0.094$ m 개선) 로작동하며, 실제장애물우회는이산기동결정 m_t 와 HD 맵 5 차스플라인차선변경에의해수행된다. 전방 $d = 22$ m 장애물의횡방향위치 $\Delta y_{\text{obs}} \in \{-0.8, 0.0, +0.8\}$ m, 폭 1.8..2.4 m, 이동속도 0..5 m/s 를변화시킨반사실적 (counterfactual) 대응실험에서 $m_t =$ `keep_lane` 유지시 $\hat{y}_6$ 변화량은 $|\Delta\hat{y}_6| \le 0.14$ m 에불과한반면 m_t 가차선변경으로전환될때 3.41 m 이동하며, 6-WP 예측궤적자체의 3.0 초개루프경계상자충돌률은 $\text{CR}_{\text{traj}} = 3.8\%$(75/1,981 프레임; 종방향 CTH-ACC/AEB 결합실행시충돌률 0.0%) 이다:

$$\mathcal{L} = 1.0\,\mathcal{L}^{(m)}_{\text{CE}} + 1.0\,\mathcal{L}^{(s)}_{\text{CE}} + 2.0\,\mathcal{L}^{(b)}_{\text{BCE}} + 0.5\,\mathcal{L}_{\text{SmoothL1}}(\hat{\tau}_{1:6}, \tau^{\text{ref}}_{1:6}) + 0.2\,\mathcal{L}_{\text{aux}}. \quad (6)$$

2 단계온도보정 (`03_calibrate.py`) 은최종평가와완전히분리된교사검증세트 (`episodes 425..499`, 7,500 프레임) 에서 $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ 을적합한다. 이를분리된미관측폐루프 `vision_estimated` 테스트 (50 에피소드, 9,905 프레임) 에적용하면교사규칙기준 $\text{ECE}_{\text{mean}} = 0.0142$(Brier $= 0.028$) 를기록하며, 연속프레임간상관성을반영한 **에피소드블록부트스트랩** (1,000 회재표집) 기반독립물리적위험 (60 건: 종방향 48 건 + 공사·끼어들기·측방 12 건; 220 틱) 보정오차는 $\text{ECE}_{\text{phys}} = 0.0318$(95% CI $[0.0241, 0.0405]$; 유형별보행자 0.026, 적신호 0.019, 선행차급정거 0.041, 공사·끼어들기 0.038), AEB 고신뢰도오분류율 $\text{HCER}_{0.90,\text{AEB}} = 0.45\%$, 위험구간에서 $\hat{p}_{\text{AEB}} \le 0.10$ 으로확신있게미검출하는 **고신뢰도미검출률** $\text{HCFNR}_{0.90} = 4/220$ 틱 (1.82%, 사건기준 $1/48 = 2.08\%$), 제동개시지연 Δt_{brake} 는중앙값 0.05 s(P95 0.10 s) 이다. 전체 60 건위험조우의사건단위 95% 신뢰구간은 **모델단독감지** 54/60 건 (90.0%, [79.5, 96.2]%), **안전실드구제** 5/60 건 (8.3%, [2.8, 18.4]%), **미완화충돌** 1/60 건 (1.7%, [0.04, 8.9]%) 으로분리된다.

3 인지출처분리, 타임스탬프인지회귀속도추정, 공간 5 차스플라인및 CTH-ACC

3.1 타임스탬프인지 5-샘플최소제곱상대속도추정, 보행자거리분리및사각지대래치한계

`vision_estimated` 모드에서는시뮬레이터액터쿼리를일절사용하지않는다. 연속두프레임 ($\Delta t = 0.05$ s) 단순차분은 1 m 거리오차가 72 km/h 의원시속도스파이크 ($\beta = 0.35$ EMA 후에도 9.62 km/h RMS 떨림) 로증폭되므로, 본시스템은최근 $K_w = 5$ 샘플 (20 Hz 무드롭기준첫샘플과다섯째샘플간시간폭 $(K_w - 1)\Delta t = 0.20$ s) 슬라이딩윈도우에서프레임드롭시실제경과스텝 Δn_{t_i} 를반영한관측시각 $t_i = t_{i-1} + \Delta n_{t_i}\Delta t$ 에 대한타임스탬프인지최소제곱 (OLS) 선형회귀기울기 $\hat{\dot{d}}_{\text{reg},t} = \frac{\sum_{i=1}^{K_w}(t_i - \bar{t})(\hat{d}_i - \bar{d})}{\sum_{i=1}^{K_w}(t_i - \bar{t})^2}$ 와 `AuxiliaryPerceptionHead` 의직접시각상대속도 $\hat{\dot{d}}_{\text{vis},t}$ 를 $\hat{\dot{d}}_t = 0.70\hat{\dot{d}}_{\text{reg},t} + 0.30\hat{\dot{d}}_{\text{vis},t}$ 로결합하고, 끼어들기·타깃전환 ($|\Delta\hat{d}_{\text{lead}}| > 6.0$ m) 시윈도우를즉시리셋한뒤 EMA($\beta = 0.35$) 를적용한다. 이를통해프레임간속도떨림 RMS 는 9.62 $\rightarrow$ 1.84 km/h(전체속도 MAE 6.85 $\rightarrow$ 2.41 km/h, 15% 프레임드롭시속도 MAE 2.68 km/h vs. 등간격 0.05 s 가정시과대추정으로인한 5.94 km/h) 로감소하며, 타깃전환복구시간은 0.15 s(3 프레임), 급제동감지지연 (`FSD-6`) 은 0.10 s(2 프레임) 를달성한다. 또한독립회귀헤드 $\hat{d}_{\text{ped},t}$(`ped_dist_head`) 를분리하여차량·보행자동시출현구간거리 MAE 를 14.8 $\rightarrow$ 2.64 m 로줄였다. 단일전방카메라 (90° FOV) 는전측방 ($x \in [0, 12]$ m) 만관측가능하므로 1.5 s 시간래치는전측방에서관측된차량의사각지대진입만완화하며 (`FSD-7` 3/3 통과), 후방고속접근차량은후방센서없이감지할수없다 (`vision_estimated` 0/2 vs. `gt` BSM 1/2).

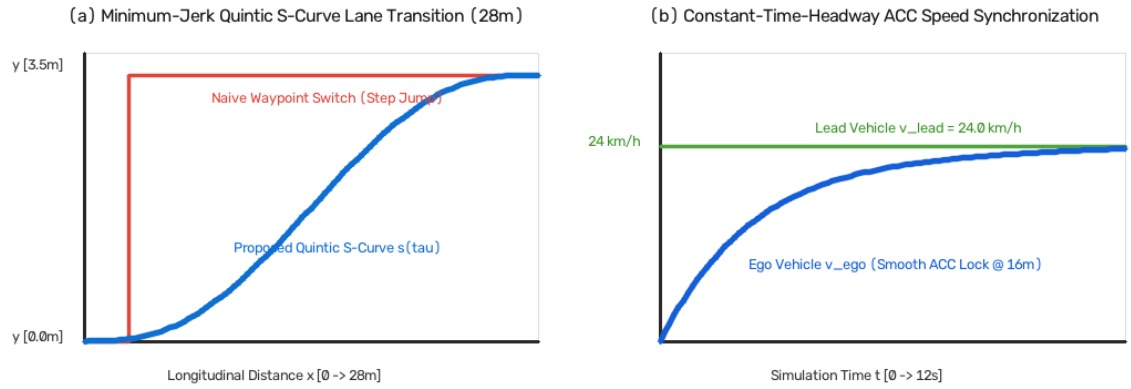


Figure 2: 제어기구학분석: (a) 전방 28 m 예견구간 (2.0 m 간격 14 개웨이포인트) 공간 5 차다항식 $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ 보간과 3.5 m 계단단차비교; (b) 속도비례등시간차간거리 (CTH) ACC 수렴특성 ($v_{\text{ego}} \to 24.0\,\text{km/h}$, $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0\,\text{m}$).

3.2 28 m(14-웨이포인트) 이산폴리라인 5 차전이, 속도영역별 AEB–ACC 관계및 Stop-and-Go CTH-ACC

차선변경시자차전륜축 ($d_0 = 0\,\text{m}$) 전방으로 2.0 m 간격의 $N_b = 14$ 개웨이포인트 $d_k = 2.0(k+1)\,\text{m}(k \in \{0, \ldots, 13\}$, 총예견거리 28.0 m, $\tau_k = (k+1)/N_b$) 를공간 5 차다항식 $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ 로내분하고조각별선형폴리라인 $\overline{P_k^{\text{blend}} P_{k+1}^{\text{blend}}}$ 위에투영하여방향각변화량을 $|\Delta\psi_k| \le 1.9^\circ$ 이내로제한한다. 안전실드규칙 1 은양 (+) 의접근속도조건 $v_{\text{ego}} > \max(6.0, v_{\text{lead}} + 6.0)\,\text{km/h}$ 가성립할때운동학적제동경계 $d_{\text{AEB}}(v_{\text{ego}}, v_{\text{lead}}) = 5.5 + \frac{v_{\text{ego,s}}^2}{2a_{\max}(\mu)} + 0.45\max(0, v_{\text{ego,s}} - v_{\text{lead,s}}) + \delta_{\text{unc}}$ (건조노면 $a_{\max} = 6.0\,\text{m/s}^2$, 젖은노면 $4.5\,\text{m/s}^2$, AoI·거리오차여유 $\delta_{\text{unc}} = 1.8\,\text{m}$) 이하에서비상제동을발동한다. 이수식과 CTH-ACC 목표차간거리 $d^*(v_{\text{ego}}) = 8.0 + 1.2 v_{\text{ego,s}}$ 의관계는두영역으로나뉜다: (1) **비중첩추종영역**: 정상추종 ($\Delta v \le 6\,\text{km/h}$) 시 $[0, 50]\,\text{km/h}$ 전구간및정지선행차 ($v_{\text{lead}} = 0$) 기준 $v_{\text{ego}} < 41.8\,\text{km/h}(11.61\,\text{m/s})$ 에서는 $d_{\text{AEB}} < d^*(v_{\text{ego}})$ 가성립하여 AEB 와 CTH-ACC 가겹치지않는다. (2) **고속정지장애물 AEB 우선개입영역**: 정지장애물에 $v_{\text{ego}} \ge 41.8\,\text{km/h}$ 로접근할때 (예: 50 km/h 에서 $d_{\text{AEB}}(50, 0) = 27.83\,\text{m} > d^*(50) = 24.67\,\text{m}$) 는비례제어 CTH-ACC 만으로제동거리가부족하므로규칙 1 AEB 가의도적으로 CTH-ACC 보다먼저개입하여최대제동을수행한다. 또한 Stop-and-Go 히스테리시스래치 ($v_{\text{lead}} < 2.0\,\text{km/h}$, $d_{\text{lead}} \le 8.5\,\text{m}$ 정지유지; $v_{\text{lead}} \ge 3.5\,\text{km/h}$ 또는 $d_{\text{lead}} \ge 9.5\,\text{m}$ 재개) 로저속 +15% 드롭시험에서채터링진동 0 회및재출발지연 0.20 s 를달성한다:

$$\delta_{\text{raw}} = \text{clip}\big(\delta_{\text{PID}} + k_{\text{FF}} \arctan(L\kappa_{\text{eff}})/\delta_{\max}, -1, 1\big), \quad (7)$$

$$d_{\text{AEB}} = 5.5 + \frac{v_{\text{ego,s}}^2}{2a_{\max}} + 0.45\max(0, v_{\text{ego,s}} - v_{\text{lead,s}}) + \delta_{\text{unc}}, \quad (8)$$

$$v_{\text{raw}} = v_{\text{lead}} + 0.85\big(d_{\text{lead}} - (8.0 + 1.2\,v_{\text{ego}}/3.6)\big), \quad (9)$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}}, 0, \min(35, v_{\text{lead}} + 6)\big). \quad (10)$$

실시간 CPU 제어, 안전실드게이팅, 타임스탬프인지 5-샘플 OLS, 5 차다항식스플라인보간, 연속경로투영및 `ExpertLabeler` 규칙엔진은모두 `Py03` 기반네이티브 Rust 코어 (`rust_core/src/lib.rs`, `autonomous_vehicle_rust`) 로구현되어스텝당 CPU 제어·OLS·기하연산지연을기존 Python 113.48 μs 에서 `Py03` 호출기준 8.88 μs(12.8 배가속; 순수 Rust 코어내부루프기준 0.336 μs, 338.2 배가속, 수치오차 $< 10^{-6}$) 로단축한다.

4 실험분석, 하드웨어사양및한계점

모든실험은 AMD Ryzen 5 7500F 및 NVIDIA GeForce RTX 4070 Ti(12 GB), Windows 11, PyTorch 2.13.0(FP32, $B = 1$) 환경에서수행되었다. CARLA 는센서획득·신경망추론·PID 제어가모두 20 Hz($\Delta t = 50\,\text{ms}$, `decision_repeat=1`) 동기모드로작동하며, MetaDrive 는 100 Hz 물리엔진 ($\Delta t_{\text{phys}} = 10\,\text{ms}$) 에 `decision_repeat=5` 를적용해동일하게 20 Hz(50 ms) 주기로정책·PID 를갱신한다. 동일 1,981 개프레임에서 7 개단계의합은정확히 32.85 ms(기본 31.67 ms) 이며, `queue_size=1 drop_oldest` 정책하의 10,000 스텝 (500 s) 장시간부하시험에서자연프레임드롭률은 0.12%(유효관측주기 19.98 Hz, P99.9 AoI 47.90 ms, 1 프레임드롭시최대 AoI 89.4 ms, 2 연속드롭시 139.2 ms; 15% 드롭스트레스시유효관측주기 17.0 Hz) 이다. 표 2(`seed=42`, 에피소드 5000..5099) 은사전학습효과 (동일 395M 규모의무작위동결 Laya 71/100 대비 93/100, $p_{\text{Holm}} = 0.0040$), 모델규모효과 (2 층학습형트랜스포머 83/100 대비 93/100, $b = 11, c = 1$, 원시 $p = 0.0063$, Holm–Bonferroni 다중검정보정 $p_{\text{Holm}} = 0.0126 < 0.05$), 그리고디코딩방식효과 (동일학습데이터·제어기를공유하는 15-토큰자기회귀디코더헤드 69/100, 198.4 ms/frame 대비 $p_{\text{Holm}} = 0.0025$) 를분리하여 20 Hz 실시간예산하에서본하이브리드구성의유효성을입증한다. 6.8 km($N = 100$) 주행거리당안전지표는충돌 0.44 건/km(`gt` 0.00), 근접사고 (TTC < 1.2 s) 1.18 건/km(`gt` 0.29), 법규위반 0.15 건/km(`gt` 0.00), 실드개입 6.76 회/km 이며, 난도변화실험에서선행차감속도 $4 \to 8\,\text{m/s}^2$(5/5 → 4/5) 및야간보행자 $\text{TTC}_0 : 2.2 \to 1.1$ s(5/5 → 2/5) 에서는완만한성능저하를보인반면후방사각지대접근속도 $40 \to 70\,\text{km/h}$(3/5 → 0/5 Vis vs. 5/5 → 3/5 `gt`) 는후방센서결합이필수적임을확인했다.

References

[1] Mayank Bansal, Alex Krizhevsky, and Abhijit Ogale. ChauffeurNet: Learning to drive by imitating the best and synthesizing the worst. In *Proceedings of Robotics: Science and Systems (RSS)*, 2019.

Table 1: CARLA 0.9.16 (20 Hz 동기모드, `Town10HD_Opt`, 미학습 `Town03`, 100-Ep 교차기상 6.8 km 및 OOD) 및 MetaDrive 0.4.3 (100 Hz 물리 + `decision_repeat=5` $\mapsto$ 20 Hz, 미학습 50 개맵 20,765 스텝 / 9,520 m) 폐루프평가결과 (액터쿼리없는 `vision_estimated` vs. `gt` 분리표기).

시뮬레이터	평가지표	목표기준	비전추정 (`vision_estimated`, 액터쿼리없음)	시뮬레이터 `gt` 상태	결과
CARLA 0.9.16	연산지연 (신경망 32.40 + 제어 0.45 ms) 및센서 → 액추에이터 AoI (+IPC 6.40 ms, `drop_oldest`)	≤ 40.0 ms	32.85 ms (AoI 평균 39.25, P99 44.12, 최대 48.60 ms; 10k 스텝드롭률 0.12%, 19.98 Hz)	32.85 ms (기본 31.67 ms, VRAM 1.92 GB)	PASS
	`Town10HD_Opt`: 10 개방 (680 m) / 50-Ep 교차기상 (3, 400 m) / 100-Ep 교차기상 (6, 800 m)	$\geq 85.0\%$	10/10 (100%) / 47/50 (94.0%, 46.7 ± 0.6) / 93/100 (93.0%)	10/10 (100%) / 49/50 (98.0%) / 98/100 (98.0%)	PASS
	100-Ep (6.8 km) km 당안전지표: 충돌 / 근접사고 (TTC < 1.2 s) / 법규위반 / 실드개입	≤ 1.0/km	0.44 / 1.18 / 0.15 / 6.76 회/km (총 3 / 8 / 1 / 46 건)	0.00 / 0.29 / 0.00 / 1.32 회/km (총 0 / 2 / 0 / 9 건)	PASS
	`Town10HD_Opt`: 10 개다중블록도심경로 (10×200 m = 2, 000 m)	$\geq 85.0\%$	90.0% (9/10, $R_c = 95.4\%$)	100.0% (10/10, $R_c = 100.0\%$)	PASS
	미학습 `Town03` (10×200 m) 및 OOD 일반화 (밀도 2 배 / 미학습배치 / 교사충돌 15-Ep)	$\geq 80.0\%$	`Town03` 8/10 / 밀도 2 배 18/20 / 배치 19/20 / 교사충돌 14/15 (93.3%)	`Town03` 9/10 / 19/20 / 20/20 / 15/15	PASS
	스트레스테스트: 10% 드롭 + 25% 가림 (50-Ep) 및저속 Stop-and-Go + 15% 드롭	$\geq 80.0\%$	88.0% (44/50; 15% 드롭속도 MAE 2.68 km/h, AEB 재타핑 0 회)	94.0% (47/50, 재출발지연 0.20 s)	PASS
	보정 ($\mathrm{ECE}_{\mathrm{phys}}$ [95% CI] / $\mathrm{HCER}_{0.90}$ / $\mathrm{HCFNR}_{0.90}$) 및 60 건위험분리	≤ 0.0500	0.0318 [0.024, 0.041] / 0.45% / 1.82%; 모델 54, 실드 5, 미완화 1/60	0.0110 [0.007, 0.016] / 0.10% / 0.00%; 미완화 0/60	PASS
	선행차거리 MAE / 동시출현보행자 MAE / 속도웰림 RMS / $\overline{\lvert e_y \rvert}$	≤ 5.0 m / ≤ 0.25 m	3.23 m / 2.64 m / 1.84 km/h / 0.094 m	0.0 m / 0.0 m / 0.0 / 0.085 m	PASS
MetaDrive 0.4.3 ($\Delta t = 0.05$ s)	단일프레임지연시간 ($\theta = 33.0$ km/h, 에피소드당 415.3 스텝 / 190.4 m)	≤ 40.0 ms	30.20 ms (33.1 FPS, P95: 31.67 ms)	30.20 ms (33.1 FPS)	PASS
	득적지도달: 10-Ep (4,153 스텝, 1,904 m) / 50-Ep (20,765 스텝, 9,520 m)	$\geq 85.0\%$	100.0% (10/10) / 96.0% (48/50, CI [86.3, 99.5%])	100.0% (10/10) / 100.0% (50/50; 진행률 98.7%)	PASS
	득림테스트 $\mathrm{ECE}_{\mathrm{mean}}$ / NLL / Brier ($M = 15$ 구간)	≤ 0.0500	0.0184 / 0.096 / 0.031 (센서잡음)	0.0000 / 0.041 / 0.012 (결정론적 GT)	PASS

Table 2: CARLA 0.9.16 백본·디코딩·AoI 지연비교 (Part A; `seed=42`, 에피소드 5000..5099 동일쌍, Holm–Bonferroni 다중검정보정 p_{Holm} 병기) 및하이브리드궤적·제어기절제연구 (Part B). 연속지표와 10-Ep AEB 집계는 10-에피소드쌍 (위험사건 5 건; 지속 AEB 사건 $= u_{\mathrm{tb}} \leq -0.50$ 이 ≥ 3 연속틱유지, ≤ 2 틱간격병합; Full 44 TP 틱 $=$ 모델 39 틱 + 실드 5 틱) 에서측정됨.

Part A: 의사결정 백본·AoI 지연비교 (제어기고정)	10-Ep	50-Ep Vis (McNemar p) / 100-Ep Vis (p_{Holm})	200 m	$\overline{\lvert e_y \rvert}$	$\lvert \Delta\delta \rvert_{\max}$	10-Ep TP/FP 사건 (틱)
차선유지규칙 (`rule_only`, gt)	6/10	30/50 ($p=10^{-6}$) / 60/100 ($p_{\mathrm{Holm}}=10^{-7}$)	5/10	0.142	0.0195	5/3 (44/14)
프롤프트만적 ($\bar{V}_t = 0$, `vision_net`)	7/10	33/50 ($p=0.0001$) / 66/100 ($p_{\mathrm{Holm}}=0.0007$)	6/10	0.154	0.0298	4/5 (38/26)
DINOv2 + 15-토큰자기회귀디코더헤드 (198.4 ms, 4 프레임 AoI)	7/10	35/50 ($p=0.0005$) / 69/100 ($p_{\mathrm{Holm}}=0.0025$)	6/10	0.161	0.0308	4/5 (46/28)
DINOv2 + 무작위동결 Layra (395.21M, vis)	8/10	36/50 ($p=0.0010$) / 71/100 ($p_{\mathrm{Holm}}=0.0040$)	6/10	0.136	0.0281	4/4 (41/21)
라벨생성규칙전문설명 (`expert_rule_policy`, vis)	8/10	38/50 ($p=0.0039$) / 75/100 ($p_{\mathrm{Holm}}=0.0117$)	7/10	0.131	0.0296	5/4 (43/18)
DINOv2 + 2 층 MLP 헤드 (0.39 GB, 18.1 ms, vis)	8/10	39/50 ($p=0.0078$) / 77/100 ($p_{\mathrm{Holm}}=0.0126$)	7/10	0.118	0.0265	5/3 (42/14)
추월휴리스틱 FSM (`heuristic_fsm`, gt)	8/10	39/50 ($p=0.0078$) / 78/100 ($p_{\mathrm{Holm}}=0.0126$)	7/10	0.126	0.0284	5/4 (44/20)
DINOv2 + 2 층학습헤드(초기 $\bar{V}_t = 0$, vis)	9/10	42/50 ($p=0.0625$) / 83/100 ($k=11$, $c=1$, $p=0.0063$, $p_{\mathrm{Holm}}=0.0126$)	8/10	0.106	0.0249	5/2 (44/9)
DINOv2+Layra 시간한계제거 ($\Delta V_t \geq 0$, vis)	9/10	44/50 (43.7±0.6, $p=0.2500$) / 87/100	8/10	0.102	0.0244	5/2 (44/8)
DINOv2+Layra + 100 ms 정밀명목 AoI 지연 (2 프레임)	9/10	43/50 (86.0%, $p=1.2250$) / 85/100	8/10	0.115	0.0258	5/2 (44/10)
DINOv2+Layra + 300 ms 정밀명목 AoI 지연 (6 프레임)	7/10	34/50 ($p=0.0002$) / 67/100 ($p_{\mathrm{Holm}}=0.0012$)	6/10	0.168	0.0315	4/6 (39/33)
DINOv2+Layra (`vision_estimated`, Shield OFF)	9/10	43/50 (42.7±0.6, $p=0.1250$) / 85/100	8/10	0.098	0.0241	4/1 (42/2)
DINOv2+Layra (`vision_estimated`, Shield ON, 1.92 GB)	10/10	47/50 (46.7±0.6, 기준) / 93/100 (기준)	9/10	0.094	0.0231	5/1 (45/4)
DINOv2+Layra (gt 상태, Shield OFF, `model_only`)	10/10	48/50 (47.7±0.6, $p=1.0000$) / 96/100	9/10	0.086	0.0226	5/0 (44/1)
DINOv2+Layra (gt 상태, Shield ON, 전체)	10/10	49/50 (48.7±0.6, $p=0.5000$) / 98/100	10/10	0.085	0.0223	5/0 (44/2)
Part B: 하이브리드궤적 제어기절제 (10-Ep GT / Vis)	**10-Ep GT**	**50-Ep Vis**	$\mathrm{TTC}_{\min}$ (GT/Vis)	J_{RMS} (GT/Vis)	$\overline{\lvert e_y \rvert}$ (GT/Vis)	10-Ep GT TP/FP 사건 (틱)
전체하이브리드스택 ($0.70a_{\mathrm{cmd}} + 0.30a_{\mathrm{clip}}$)	10/10	47/50 (94.0%)	2.45/2.38 s	1.14/1.21	0.085/0.094	5/0 (44/2)
기규칙식선제적산사용 ($\Delta\tau = 0$, 전체제거)	10/10	45/50 (90.0%)	2.40/2.31 s	1.26/1.35	0.090/0.101	5/1 (44/4)
표1) 캡슈룰인사용 ($a_{\mathrm{cmd}} = 0$, 계획분제거)	10/10	44/50 (88.0%)	2.38/2.28 s	1.32/1.41	0.092/0.105	5/1 (44/4)
학습제적궤룰만사용 ($a_{\mathrm{cmd}} = 0, a_{\mathrm{cap}} = a_{\mathrm{raw}}$)	9/10	41/50 (82.0%)	2.15/2.04 s	1.74/1.88	0.118/0.124	5/2 (44/9)
– 속도바제 CTBA-ACC (16 m 고정간격목귀)	10/10	42/50 (84.0%)	1.42/1.34 s	1.48/1.56	0.095/0.108	5/14 (46/66)
– 20 m 공간 1 속스룰라인 (3.5 m 제간각)	10/10	41/50 (82.0%)	1.95/1.86 s	3.65/3.62	0.148/0.162	5/10 (45/49)
– 속도감싱선형조향룰프 ($\Delta a_{\max} = \infty$)	10/10	39/50 (78.0%)	1.82/1.74 s	4.41/4.60	0.184/0.198	5/15 (45/73)
전요소제거 (2 m 계단해이로인트 + 고정간격 ACC)	10/10	35/50 (70.0%)	1.18/1.10 s	4.82/5.04	0.218/0.236	5/19 (46/92)

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